Riasad Rahman Rakib



Supply Chain & Operations Analyst

Rajshahi, Bangladesh | +880-1640590915
riasadrahmanrakib.24@gmail.com | linkedin.com/in/riasad-rahman-rakib

SUMMARY

Final-year Industrial & Production Engineering student at RUET with BAT,Bangladesh factory training in downtime and process-efficiency analysis. Six Sigma White Belt and Green Belt certified. Built a MILP waste-collection routing model with 10.76%–17.52% modeled cost savings. Finalist at Efficiency Pioneer and runner-up at IPEsphere. Python, SQL and Excel for supply chain analytics.

EDUCATION

BSc in Industrial & Production Engineering (IPE), RUET — Dec 2022 – Expected Feb 2027
Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh | CGPA: 3.67 / 4.00
Relevant coursework: Operations Research, Operations Management (EOQ and cycle inventory), Industrial & Business Management, Probability & Statistics, Engineering Economics, Quality Control Management

INDUSTRIAL EXPERIENCE

Industrial Trainee, British American Tobacco Bangladesh — Feb 2026 – Mar 2026
Primary Manufacturing Department, Savar Factory

- Mapped the Lamina and Stem processing lines with Value Stream Mapping (VSM) and calculated Process Cycle Efficiency (PCE): 60.3% for Lamina and 49.8% for Stem.
- Analyzed processing capacity from batch sizes and cycle times to investigate the throughput imbalance between the Lamina and Stem lines.
- Applied Pareto analysis to downtime across nine key equipment units and identified 357 recorded stoppage events linked to full silos and CRS-bin full/not-ready conditions.
- Recommended improvements in line scheduling, buffer monitoring, bulking time and critical spare-parts readiness.

CERTIFICATIONS

Six Sigma Green Belt | **Six Sigma White Belt**

SELECTED PROJECTS & RESEARCH

A Two-Stage Clustered MILP Framework for Efficient and Resilient Waste Collection Routing, Rajshahi City Corporation

- Used K-Means clustering to split 20 Secondary Transfer Stations into four clusters, then built a MILP model in Gurobi to route a mixed fleet (1.5-ton and 3-ton sweepers, 5-ton shuttles) under capacity, time-window, multi-trip and eight-hour shift constraints; cost objective covered fuel, labor and carbon penalties.
- Modeled savings of 10.76%–17.52% versus fixed-route dispatch across 11 demand scenarios (about BDT 12,600–20,500 per day; about BDT 14,500 in the baseline), with sensitivity analysis on fuel price, traffic congestion and carbon penalty. These are research-model results, not realized savings.

Undergraduate Thesis: Resilient and Sustainable Supply Chain Network Design, Bangladesh RMG Sector — In Progress

- Building a Temporal Fusion Transformer for multi-horizon export demand forecasting and an LLM-RAG pipeline that extracts resilience and sustainability parameters from BGMEA, ILO and buyer documents.
- Designing a multi-objective MILP network model that balances cost, carbon emissions, resilience and social sustainability, using the augmented epsilon-constraint method.

COMPETITIONS & ACHIEVEMENTS

- **Finalist**, Efficiency Pioneer
- **Semi-finalist**, Powerpreneur 3.0
- **2nd Place**, IPEsphere

SKILLS

Operations & process analysis: Value Stream Mapping, Process Cycle Efficiency, Pareto analysis, downtime analysis, capacity analysis, process improvement, Six Sigma (White Belt, Green Belt)
Supply chain & planning: Supply chain network design, demand forecasting, inventory models (EOQ), Vehicle Routing Problem (VRP), cost optimization, scenario and sensitivity analysis
Optimization & modeling: Mixed-Integer Linear Programming (MILP), multi-objective optimization, K-Means clustering, Gurobi
Data & software: Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL (MySQL), Microsoft Excel (PivotTables, XLOOKUP, SUMIFS, data cleaning, charts), Microsoft Office, Jupyter Notebook

LEADERSHIP & ACTIVITIES

Assistant Secretary (Sponsorship), RUET IPE Club — 2024 – 2026

- Handled communication with sponsors and partners for club events, managed the club's Facebook page, and supported departmental events including Presentio 3.0, Mastering Communications, CaseCation and EntrePret.

Community Manager, Engineer's Diary, RUET — Jun 2023 – Dec 2023

- Scheduled content, organized podcasts and coordinated team meetings.

Languages: English (proficient), Bangla (native)